

MAVERICK HOZIEL

☎ (514)-943-1493 ✉ maverick.hoziel@mail.mcgill.ca 💻 [Maverick Hoziel](#) 🌐 hozielmaverick.github.io 📍 Montréal, QC

Profile

- Hands-on laboratory experience with high-speed IMDD transmission systems and **TFLN photonics**
- Experience with closed-loop control, including **PID tuning** and **Automatic Bias Control (ABC)**
- Skilled with **temperature-controlled testing**
- Practical experience with through-hole soldering, wire-to-wire soldering, and **electrical wiring**.
- Working knowledge of photonics modeling using **VPIphotonics**
- Two-time McGill **Varsity Lacrosse Team Captain** and Varsity Council Representative
- Fully bilingual in **French** and **English**

Education

McGill University – Montréal, QC

January 2027 – April 2028

Incoming MSc in Electrical Engineering, supervised by Professor David Plant

McGill University – Montréal, QC

August 2022 – December 2026

*Bachelor of Engineering in **Electrical Engineering**, Minor in **Aerospace Engineering** – **Internship Program***

Professional Experience

HyperLight Corporation

May 2026 – August 2026

Photonics Engineer Intern

Cambridge, MA

- Developed multiple automated test stations to characterize production and R&D datacom PICs across the extended temperature range defined by the OSFP Module Thermal Requirements (-5°C to 85°C).
- Presented weekly test results and data analysis to the head of engineering and cross-functional teams.
- Programmed, assembled, and integrated an automatic bias-control board.
- Modeled various PIC components using VPIphotonics.

Électricité Kingston

May 2023 – August 2025

Electrical Engineering Intern

Terrebonne, QC

- Managed construction projects from planning through completion, ensuring on-time delivery
- Contributed to the development of an internal AI tool designed to streamline company operations

Projects

Optical Datacom Capstone | 50+ Gbps IMDD LN/TFLN MZM

January 2026 – March 2026

- Designed, built, and tested a complete **52 Gbps IMDD LN MZM transceiver system** over 2-km and 5-km fiber links, integrating RF driver electronics and validating performance through DCA eye-diagram measurements including jitter and ER.
- Simulated **TFLN** photonic devices including **waveguide crossings**; extracted insertion loss, crosstalk, and reflection metrics.
- Compared electro-optic material behavior between bulk LN and TFLN, focusing on modulation efficiency (V_π), bandwidth potential, and reduced power consumption.

AA-1C Yankee Electric Propulsion Conversion | Avionics Architecture, Power BI

December 2025

- Redesigned the aircraft's electrical and avionics architecture for electric propulsion, defining system interfaces and control signal paths between cockpit controls, inverter, contactor module, BMS, and avionics loads.
- Integrated CAN communication between the BMS, inverter, and cockpit displays to manage system status and safety related signals.

Leadership

Captain, McGill Redbirds Men's Lacrosse Team

2024 – Present

- **Elected for a second season as team captain; led team culture and represented athletes on the Varsity Council.**

Recipient, Evans Huber Memorial Award – Most Dedicated Player

2023 – 2025

- **First McGill lacrosse player to receive the award three consecutive years.**